CHAPTER 7

Fractions

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples; write your own.

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Figure it Out · 7.1 Fractional Units and Equal Shares

1 **Three guavas together weigh 1 kg. If they are roughly of the same size, each guava will roughly weigh ____ kg.**

✔ ANSWER

1 kg is shared equally by 3 guavas.

1 ÷ 3 = 1/3

Each guava weighs **1/3 kg**.

2 **A wholesale merchant packed 1 kg of rice in four packets of equal weight. The weight of each packet is ___ kg.**

✔ ANSWER

1 ÷ 4 = 1/4

Each packet weighs **1/4 kg**.

3 **Four friends ordered 3 glasses of sugarcane juice and shared it equally among themselves. Each one drank ____ glass of sugarcane juice.**

✔ ANSWER

3 glasses shared by 4 friends: 3 ÷ 4 = 3/4

Each one drank **3/4 glass**.

4 **The big fish weighs 1/2 kg. The small one weighs 1/4 kg. Together they weigh ____ kg.**

✔ **ANSWER**

1/2 = 1/4 + 1/4

So 1/2 + 1/4 = 1/4 + 1/4 + 1/4 = 3/4

Together they weigh **3/4 kg**.

Discuss **Find out and discuss the words for fractions used in the languages spoken in your home, city or state (one and a half, three quarters, one and a quarter, half, quarter, two and a half).**

💡 **SAMPLE ANSWER**

Example (Hindi):

Fraction	Hindi word
1½ (one and a half)	डेढ़ (dedh)
¾ (three quarters)	पौना / तीन पाव (pauna / teen paav)
1¼ (one and a quarter)	सवा (sawa)
½ (half)	आधा (aadha)
¼ (quarter)	पाव / चौथाई (paav / chauthai)
2½ (two and a half)	ढाई (dhai)

Ask your family for the words in your own language.

5 **Arrange these fraction words in order of size from the smallest to the biggest: one and a half, three quarters, one and a quarter, half, quarter, two and a half.**

✔ ANSWER

In numbers: quarter = 1/4, half = 1/2, three quarters = 3/4, one and a quarter = 1¼, one and a half = 1½, two and a half = 2½.

Quarter < half < three quarters < one and a quarter < one and a half < two and a half

⭐ 7.2 Fractional Units as Parts of a Whole

? **By dividing the whole chikki into 6 equal parts in different ways, we get 1/6 chikki pieces of different shapes. Are they of the same size?**

✔ ANSWER

Yes. Each piece is 1/6 of the same whole chikki, so each has the same amount of chikki, even though the shapes are different.

Figure it Out **The figures show different fractional units of a whole chikki. How much of a whole chikki is each piece? (a – h)**

✔ ANSWER

Count how many such equal pieces make the whole chikki.

Figure	Piece
a	**1/12**
b	**1/4**
c	**1/8**
d	**1/6**
e	**1/8**
f	**1/6**
g	**1/24**
h	**1/24**

(Check the picture in your book.)

⭐ 7.3 Measuring Using Fractional Units

? **Fold a paper strip (1 unit long) into two equal parts. What are the lengths of the two new parts?**

✔ ANSWER

Each part is **1/2 unit** long.

Fill in **Fold the strip once more to get 8 equal parts. Fill in the blank boxes: 2 times 1/8, 4 times 1/8, 6 times 1/8, 8 times 1/8.**

✔ ANSWER

- 1 time 1/8 = **1/8**
- 2 times 1/8 = **2/8** = 1/4
- 3 times 1/8 = **3/8**
- 4 times 1/8 = **4/8** = 1/2

- 5 times 1/8 = **5/8**
- 6 times 1/8 = **6/8** = 3/4
- 7 times 1/8 = **7/8**
- 8 times 1/8 = **8/8** = 1

Figure it Out · 7.3 Measuring Using Fractional Units

1 **Continue this table of 1/2 for 2 more steps. (The table goes up to 1/2 + 1/2 + 1/2 + 1/2 + 1/2 = 5 times half.)**

✔ ANSWER

- 1/2 + 1/2 + 1/2 + 1/2 + 1/2 + 1/2 = **6 times half** = 6/2 = 3
- 1/2 + 1/2 + 1/2 + 1/2 + 1/2 + 1/2 + 1/2 = **7 times half** = 7/2

2 **Can you create a similar table for 1/4?**

✔ ANSWER

Yes:

- 1/4 = 1 time quarter
- 1/4 + 1/4 = 2 times quarter = 2/4
- 1/4 + 1/4 + 1/4 = 3 times quarter = 3/4
- 1/4 + 1/4 + 1/4 + 1/4 = 4 times quarter = 4/4 = 1
- 1/4 + 1/4 + 1/4 + 1/4 + 1/4 = 5 times quarter = 5/4

3 **Make 1/3 using a paper strip. Can you use this to also make 1/6?**

✔ ANSWER

Fold the strip into 3 equal parts (fold one end over, then the other end over it, so all 3 layers match). Each part is **1/3**.

Now fold the folded strip into half. Each of the 3 parts becomes 2 parts, so there are 6 equal parts.

Yes, each small part is **1/6**.

4 (a) **Draw a picture and write an addition statement to show: 5 times 1/4 of a roti.**

✔ ANSWER

Picture: 1 whole roti cut into 4 quarters, and 1 more quarter piece (5 quarter pieces).

1/4 + 1/4 + 1/4 + 1/4 + 1/4 = **5/4** = 5 times quarter = 1 + 1/4

4 (b) **Draw a picture and write an addition statement to show: 9 times 1/4 of a roti.**

✔ ANSWER

Picture: 2 whole rotis cut into quarters (8 pieces) and 1 more quarter piece (9 quarter pieces).

1/4 + 1/4 + 1/4 + 1/4 + 1/4 + 1/4 + 1/4 + 1/4 + 1/4 = **9/4** = 9 times quarter = 2 + 1/4

5 **Match each fractional unit (1/3, 1/5, 1/8, 1/6) with the correct picture.**

💡 **SAMPLE ANSWER**

Count the equal parts the whole shape is divided into:

- Shape divided into 3 equal parts → **1/3**
- Shape divided into 5 equal parts → **1/5**
- Shape divided into 6 equal parts → **1/6**
- Shape divided into 8 equal parts → **1/8**

(Check the picture in your book.)

⭐ 7.4 Marking Fraction Lengths on the Number Line

1 **Here, a length of 1 unit is divided into three equal parts. Write the fraction that gives the length of the blue line.**

✔ **ANSWER**

Each part is 1/3. The blue line covers 2 parts.

Length = **2/3** (check the picture in your book).

2 **Here, a unit is divided into 5 equal parts. Write the fractions that give the lengths of the blue lines.**

✔ **ANSWER**

Each part is 1/5.

The blue lines are **2/5** and **4/5** long (check the picture in your book).

3 **Now, a unit is divided into 8 equal parts. Write the appropriate fractions.**

✔ ANSWER

Each part is 1/8. The marks from 0 to 1 are:

1/8, 2/8, 3/8, 4/8, 5/8, 6/8, 7/8, 8/8 (= 1)

Figure it Out · 7.4 Marking Fraction Lengths on the Number Line

1 **On a number line, draw lines of lengths 1/10, 3/10 and 4/5.**

✔ ANSWER

- Draw a number line and mark 0 and 1.
- Divide 0 to 1 into 10 equal parts. Each part is 1/10.
- 1/10: line from 0 to the 1st mark.
- 3/10: line from 0 to the 3rd mark.
- 4/5 = 8/10: line from 0 to the 8th mark (or divide 0–1 into 5 equal parts and go to the 4th mark).

2 **Write five more fractions of your choice and mark them on the number line.**

💡 SAMPLE ANSWER

Example: 1/4, 3/4, 2/7, 5/6, 3/2.

- 1/4, 3/4: divide 0–1 into 4 equal parts; mark the 1st and 3rd marks.
- 2/7: divide 0–1 into 7 parts; mark the 2nd mark.
- 5/6: divide 0–1 into 6 parts; mark the 5th mark.
- 3/2: divide each unit into 2 parts; mark the 3rd half mark (between 1 and 2).

3 **How many fractions lie between 0 and 1? Think, discuss with your classmates, and write your answer.**

✔ ANSWER

Infinitely many (countless) fractions.

We can always divide the unit into more equal parts (1/2, 1/3, 1/4, ... 1/100, 1/1000, ...), so we never run out of fractions between 0 and 1.

4 **Each part between 0 and 1 is 1/2. The blue line is 1/2 unit long. Write the fraction that gives the length of the black line.**

✔ ANSWER

The black line covers 3 halves (from 0 to between 1 and 2).

Length = **3/2** (check the picture in your book).

5 **The unit is divided into 5 equal parts. Write the fractions that give the lengths of the**

black lines.

✔ ANSWER

The black lines go beyond 1, to the marks after 5/5.

Lengths: **6/5, 7/5, 8/5, 9/5** (check the picture in your book).

⭐ 7.5 Mixed Fractions

? **Write down all the fractions you marked on the number line earlier and classify them: lengths less than 1 unit and lengths more than 1 unit.**

💡 SAMPLE ANSWER

Less than 1 unit	More than 1 unit
1/2, 2/3, 2/5, 4/5, 1/8 ... 7/8, 1/10, 3/10	3/2, 6/5, 7/5, 8/5, 9/5

? **Did you notice something common between the fractions that are greater than 1?**

✔ ANSWER

In fractions greater than 1, the numerator is larger than the denominator. In fractions less than 1, the numerator is smaller.

Figure it Out · 7.5 Mixed Fractions (whole units)

1 How many whole units are there in 7/2?

✔ ANSWER

7/2 = 1/2 + 1/2 + 1/2 + 1/2 + 1/2 + 1/2 + 1/2

Each 2 halves make 1. 7 halves = 3 pairs + 1 half.

7/2 = 3 + 1/2, so there are **3 whole units**.

2 How many whole units are there in 4/3 and in 7/3?

✔ ANSWER

3 thirds make 1.

- 4/3 = 1 + 1/3 → **1 whole unit**
- 7/3 = 2 + 1/3 → **2 whole units**

Figure it Out · 7.5 Writing fractions greater than one as mixed numbers

1 Figure out the number of whole units in each of the following fractions: a. 8/3 b. 11/5 c. 9/4

✔ ANSWER

- a. 8/3 = 2 + 2/3 → **2**
- b. 11/5 = 2 + 1/5 → **2**
- c. 9/4 = 2 + 1/4 → **2**

2 **Can all fractions greater than 1 be written as such mixed numbers?**

✔ ANSWER

Yes, except those that are exactly whole numbers.

For example, 8/4 = 2 has no fractional part left over, so it is just a whole number, not a mixed number.

All other fractions greater than 1 can be written as a whole part + a fraction less than 1.

3 **Write the following fractions as mixed fractions (e.g., 9/2 = 4½): a. 9/2 b. 9/5 c. 21/19 d. 47/9 e. 12/11 f. 19/6**

✔ ANSWER

Divide the numerator by the denominator: quotient = whole part, remainder = new numerator.

- a. 9 ÷ 2 = 4 remainder 1 → **4 1/2**
- b. 9 ÷ 5 = 1 remainder 4 → **1 4/5**
- c. 21 ÷ 19 = 1 remainder 2 → **1 2/19**
- d. 47 ÷ 9 = 5 remainder 2 → **5 2/9**
- e. 12 ÷ 11 = 1 remainder 1 → **1 1/11**
- f. 19 ÷ 6 = 3 remainder 1 → **3 1/6**

Figure it Out · 7.5 Mixed numbers as fractions

a **Write 3 1/4 as a fraction.**

✔ ANSWER

Each whole = 4/4. 3 wholes = 3 × 4 = 12 quarters.

12/4 + 1/4 = **13/4**

b **Write 7 2/3 as a fraction.**

✔ ANSWER

7 × 3 = 21 thirds.

21/3 + 2/3 = **23/3**

c **Write 9 4/9 as a fraction.**

✔ ANSWER

9 × 9 = 81 ninths.

81/9 + 4/9 = **85/9**

d **Write 3 1/6 as a fraction.**

✔ ANSWER

3 × 6 = 18 sixths.

18/6 + 1/6 = **19/6**

e **Write 2 3/11 as a fraction.**

✔ ANSWER

2 × 11 = 22 elevenths.

22/11 + 3/11 = **25/11**

f **Write 3 9/10 as a fraction.**

✔ ANSWER

3 × 10 = 30 tenths.

30/10 + 9/10 = **39/10**

⭐ 7.6 Equivalent Fractions · Fraction wall

? **Are the lengths 1/2 and 2/4 equal? Are the lengths 2/4 and 4/8 equal?**

✔ ANSWER

Yes, both pairs are equal. So 1/2 = 2/4 = 4/8.

? **Check whether 1/3 and 2/6 are equivalent fractions or not, using paper strips.**

✔ ANSWER

Fold one strip into 3 parts and another into 6 parts. 2 pieces of 1/6 exactly cover 1 piece of 1/3.

Yes, 1/3 = 2/6.

1 **Are the lengths 1/2 and 3/6 equal?**

✔ ANSWER

In the fraction wall, 3 pieces of 1/6 cover exactly 1/2.

Yes, 1/2 = 3/6.

2 **Are 2/3 and 4/6 equivalent fractions? Why?**

✔ ANSWER

Yes. In the fraction wall, 4 pieces of 1/6 have exactly the same length as 2 pieces of 1/3.

Also 2/3 = (2 × 2)/(3 × 2) = 4/6.

3 **How many pieces of length 1/6 will make a length of 1/2?**

✔ ANSWER

1/2 = 3/6

3 pieces

4 **How many pieces of length 1/6 will make a length of 1/3?**

✔ ANSWER

1/3 = 2/6

2 pieces

Figure it Out · 7.6 Equivalent Fractions (fraction wall)

1 **Are 3/6, 4/8, 5/10 equivalent fractions? Why?**

✔ ANSWER

Yes. In the fraction wall, each of them has the same length as 1/2.

3/6 = 1/2, 4/8 = 1/2, 5/10 = 1/2 (the numerator is half the denominator in each).

2 **Write two equivalent fractions for 2/6.**

✔ ANSWER

2/6 = (2 ÷ 2)/(6 ÷ 2) = 1/3

2/6 = (2 × 2)/(6 × 2) = 4/12

1/3 and 4/12 (others: 3/9, 6/18 ...)

3 **4/6 = ___ = ___ = ___ = ___ (Write as many as you can)**

✔ ANSWER

Divide or multiply the numerator and denominator by the same number:

4/6 = **2/3 = 6/9 = 8/12 = 10/15 = 12/18 = 14/21 = ...**

Figure it Out · 7.6 Understanding equivalent fractions using equal shares

1 **Three rotis are shared equally by four children. Show the division in the picture and**

write a fraction for how much each child gets. Also write the division, addition and multiplication facts.

✔ ANSWER

Picture: cut each of the 3 rotis into 4 equal pieces; each child takes 1 piece from each roti (3 pieces of 1/4).

Fraction each child gets: **3/4 roti**

Division fact: **3 ÷ 4 = 3/4**

Addition fact: **3 = 3/4 + 3/4 + 3/4 + 3/4**

Multiplication fact: **3 = 4 × 3/4**

2 **Draw a picture to show how much each child gets when 2 rotis are shared equally by 4 children. Also write the division, addition and multiplication facts.**

✔ ANSWER

Picture: cut each of the 2 rotis into 2 halves; each of the 4 children gets 1 half.

Each child gets **2/4 = 1/2 roti**.

Division fact: **2 ÷ 4 = 2/4 = 1/2**

Addition fact: **2 = 1/2 + 1/2 + 1/2 + 1/2**

Multiplication fact: **2 = 4 × 1/2**

3 **Anil was in a group where 2 cakes were divided equally among 5 children. How much cake would Anil get?**

✔ ANSWER

2 ÷ 5 = 2/5

Anil gets **2/5 of a cake**.

? **If there are 10 children in my group, how many cakes will I need so that they get the same amount of cake as Anil?**

✔ ANSWER

10 children is 2 times 5 children, so we need 2 times 2 cakes.

4 cakes. (4/10 = 2/5)

⭐ 7.6 Equivalent Fractions · Equal shares (in-text)

? **Find some more fractions equivalent to 1/2.**

✔ ANSWER

4/8, 5/10, 6/12, 10/20, 50/100 ... (the denominator is 2 times the numerator).

? **Equally divide the rotis and write down the share of each child: 2 rotis among 3 children, 4 rotis among 6 children, 6 rotis among 9 children. Are the shares the same? Why?**

✔ ANSWER

Shares: 2/3, 4/6 and 6/9.

Yes, the shares are the same.

4/6 = (2 × 2)/(3 × 2) = 2/3 and 6/9 = (2 × 3)/(3 × 3) = 2/3.

The rotis and the children are both doubled or tripled, so each child gets the same amount.

? **Do you notice anything about the relationship between the numerator and denominator in each of these fractions (2/3, 4/6, 6/9)?**

✔ ANSWER

Each fraction is made from 2/3 by multiplying the numerator and the denominator by the same number (2/3 → ×2 → 4/6, ×3 → 6/9).

Also, the denominator is always 1½ times the numerator (e.g., 4 × 1½ = 6).

Figure it Out · 7.6 Find the missing numbers

a **5 glasses of juice shared equally among 4 friends is the same as ____ glasses shared equally among 8 friends. So, 5/4 = ?/8**

✔ ANSWER

8 friends = 2 × 4 friends, so glasses = 2 × 5 = 10.

10 glasses; 5/4 = **10/8**

b **4 kg of potatoes divided equally in 3 bags is the same as 12 kg of potatoes divided equally in ___ bags. So, 4/3 = 12/?**

✔ ANSWER

12 kg = 3 × 4 kg, so bags = 3 × 3 = 9.

9 bags; 4/3 = **12/9**

c **7 rotis divided among 5 children is the same as ____ rotis divided among ____ children. So, 7/5 = ?**

✔ ANSWER

Multiply both by the same number (here 2).

14 rotis among 10 children; 7/5 = **14/10**

(others: 21/15, 28/20 ...)

⭐ 7.6 Comparing shares (in-text)

? **Suppose the number of children is kept the same, but the number of units being shared is increased. What can you say about each child's share now? Why? Discuss how this explains 1/5 < 2/5, 3/7 < 4/7 and 1/2 < 5/8.**

✔ ANSWER

Each child's share becomes larger, because more units are shared among the same number of people.

- 1/5 < 2/5: 5 children share 1 unit vs 2 units.
- 3/7 < 4/7: 7 children share 3 units vs 4 units.
- 1/2 = 4/8, and 8 children sharing 4 units get less than 8 children sharing 5 units, so 1/2 < 5/8.

1 **Group 1: 3 glasses of sugarcane juice divided equally among 4 children. Group 2: 7 glasses divided equally among 10 children. In which group does each child get a larger share?**

✔ ANSWER

Group 1: 3/4; Group 2: 7/10.

Make the denominators the same: 3/4 = 15/20, 7/10 = 14/20.

15/20 > 14/20, so 3/4 > 7/10.

Group 1 children get a larger share.

2 **Group 1: 4 glasses of sugarcane juice divided equally among 7 children. Group 2: 5 glasses divided equally among 7 children. In which group does each child get a larger share?**

✔ ANSWER

Group 1: 4/7; Group 2: 5/7.

Same number of children, more juice in Group 2: 5/7 > 4/7.

Group 2 children get a larger share.

? **Which groups were easier to compare? Why?**

✔ ANSWER

The groups in question 2 were easier, because the number of children (the denominator) was the same, so we only had to compare the numerators.

⭐ 7.6 Equivalent fractions with the same fractional unit

a **Find equivalent fractions with the same fractional unit: 7/2 and 3/5**

✔ ANSWER

LCM of 2 and 5 is 10.

7/2 = 35/10, 3/5 = 6/10

35/10 and 6/10

b **Find equivalent fractions with the same fractional unit: 8/3 and 5/6**

✔ ANSWER

6 is a multiple of 3.

8/3 = 16/6, 5/6 stays 5/6

16/6 and 5/6

c **Find equivalent fractions with the same fractional unit: 3/4 and 3/5**

✔ ANSWER

LCM of 4 and 5 is 20.

3/4 = 15/20, 3/5 = 12/20

15/20 and 12/20

d **Find equivalent fractions with the same fractional unit: 6/7 and 8/5**

✔ ANSWER

LCM of 7 and 5 is 35.

6/7 = 30/35, 8/5 = 56/35

30/35 and 56/35

e **Find equivalent fractions with the same fractional unit: 9/4 and 5/2**

✔ ANSWER

4 is a multiple of 2.

9/4 stays 9/4, 5/2 = 10/4

9/4 and 10/4

f **Find equivalent fractions with the same fractional unit: 1/10 and 2/9**

✔ ANSWER

LCM of 10 and 9 is 90.

1/10 = 9/90, 2/9 = 20/90

9/90 and 20/90

g **Find equivalent fractions with the same fractional unit: 8/3 and 11/4**

✔ ANSWER

LCM of 3 and 4 is 12.

8/3 = 32/12, 11/4 = 33/12

32/12 and 33/12

h **Find equivalent fractions with the same fractional unit: 13/6 and 1/9**

✔ ANSWER

LCM of 6 and 9 is 18.

13/6 = 39/18, 1/9 = 2/18

39/18 and 2/18

Figure it Out · 7.6 Expressing a fraction in lowest terms

a Express in lowest terms: 17/51

✔ ANSWER

17 × 3 = 51, so HCF = 17.

17 ÷ 17 = 1, 51 ÷ 17 = 3

1/3

b Express in lowest terms: 64/144

✔ ANSWER

HCF of 64 and 144 is 16.

64 ÷ 16 = 4, 144 ÷ 16 = 9

4/9

c Express in lowest terms: 126/147

✔ ANSWER

HCF of 126 and 147 is 21.

126 ÷ 21 = 6, 147 ÷ 21 = 7

6/7

d Express in lowest terms: 525/112

✔ ANSWER

HCF of 525 and 112 is 7.

525 ÷ 7 = 75, 112 ÷ 7 = 16

75/16 (= 4 11/16)

Figure it Out · 7.7 Comparing Fractions

1 (a) **Compare and justify: 8/3, 5/2**

✔ ANSWER

Common denominator 6: 8/3 = 16/6, 5/2 = 15/6

16/6 > 15/6, so **8/3 > 5/2**

1 (b) **Compare and justify: 4/9, 3/7**

✔ ANSWER

Common denominator 63: 4/9 = 28/63, 3/7 = 27/63

28/63 > 27/63, so **4/9 > 3/7**

1 (c) **Compare and justify: 7/10, 9/14**

✔ ANSWER

Common denominator 70: 7/10 = 49/70, 9/14 = 45/70

49/70 > 45/70, so **7/10 > 9/14**

1 (d) **Compare and justify: 12/5, 8/5**

✔ ANSWER

Same denominator, compare numerators: 12 > 8

12/5 > 8/5

1 (e) **Compare and justify: 9/4, 5/2**

✔ ANSWER

5/2 = 10/4

9/4 < 10/4, so **9/4 < 5/2**

2 (a) **Write in ascending order: 7/10, 11/15, 2/5**

✔ ANSWER

LCM of 10, 15, 5 = 30

7/10 = 21/30, 11/15 = 22/30, 2/5 = 12/30

12 < 21 < 22

2/5 < 7/10 < 11/15

2 (b) **Write in ascending order: 19/24, 5/6, 7/12**

✔ ANSWER

LCM of 24, 6, 12 = 24

19/24, 5/6 = 20/24, 7/12 = 14/24

14 < 19 < 20

7/12 < 19/24 < 5/6

3 (a) **Write in descending order: 25/16, 7/8, 13/4, 17/32**

✔ ANSWER

LCM of 16, 8, 4, 32 = 32

25/16 = 50/32, 7/8 = 28/32, 13/4 = 104/32, 17/32

104 > 50 > 28 > 17

13/4 > 25/16 > 7/8 > 17/32

3 (b) **Write in descending order: 3/4, 12/5, 7/12, 5/4**

✔ ANSWER

LCM of 4, 5, 12 = 60

3/4 = 45/60, 12/5 = 144/60, 7/12 = 35/60, 5/4 = 75/60

144 > 75 > 45 > 35

12/5 > 5/4 > 3/4 > 7/12

⭐ 7.8 Addition and Subtraction of Fractions (in-text)

? **Meena and her brother ate 3/4 of the chikki. How much of the total chikki is remaining?**

✔ ANSWER

Whole chikki = 4/4

4/4 − 3/4 = **1/4** of the chikki is remaining.

? **Try adding 4/7 + 6/7 using a number line. Do you get the same answer?**

✔ ANSWER

Divide each unit into 7 equal parts. Start at 0, move 4 parts to 4/7, then 6 more parts.

We reach the 10th mark = 10/7 = 1 3/7.

Yes, the same answer: 10/7 = 1 3/7.

? **Try doing 6/7 − 4/7 using the number line.**

✔ ANSWER

Divide 0–1 into 7 equal parts. Start at 6/7 and move 4 parts back.

We reach **2/7**, the same answer.

? **For 3/4 − 2/3: why did we multiply the numerator and denominator of 3/4 by 3, and of 2/3 by 4?**

✔ ANSWER

We want the same denominator for both. 12 is a common multiple of 4 and 3.

4 × 3 = 12, so 3/4 is multiplied by 3/3; 3 × 4 = 12, so 2/3 is multiplied by 4/4.

Multiplying both top and bottom by the same number gives an **equivalent fraction** (value does not change).

Figure it Out · 7.8 Adding fractions

1 (a) **Add using Brahmagupta's method: 2/7 + 5/7 + 6/7**

✔ ANSWER

Same denominator: (2 + 5 + 6)/7

= **13/7 = 1 6/7**

1 (b) **Add using Brahmagupta's method: 3/4 + 1/3**

✔ ANSWER

LCM = 12: 9/12 + 4/12

= **13/12 = 1 1/12**

1 (c) **Add using Brahmagupta's method: 2/3 + 5/6**

✔ ANSWER

LCM = 6: 4/6 + 5/6 = 9/6

= **3/2 = 1 1/2**

1 (d) **Add using Brahmagupta's method: 2/3 + 2/7**

✔ ANSWER

LCM = 21: 14/21 + 6/21

= **20/21**

1 (e) **Add using Brahmagupta's method: 3/4 + 1/3 + 1/5**

✔ ANSWER

LCM = 60: 45/60 + 20/60 + 12/60

= **77/60 = 1 17/60**

1 (f) **Add using Brahmagupta's method: 2/3 + 4/5**

✔ ANSWER

LCM = 15: 10/15 + 12/15

= **22/15 = 1 7/15**

1 (g) **Add using Brahmagupta's method: 4/5 + 2/3**

✔ ANSWER

LCM = 15: 12/15 + 10/15

= **22/15 = 1 7/15**

1 (h) **Add using Brahmagupta's method: 3/5 + 5/8**

✔ ANSWER

LCM = 40: 24/40 + 25/40

= **49/40 = 1 9/40**

1 (i) **Add using Brahmagupta's method: 9/2 + 5/4**

✔ ANSWER

LCM = 4: 18/4 + 5/4

= **23/4 = 5 3/4**

1 (j) **Add using Brahmagupta's method: 8/3 + 2/7**

✔ ANSWER

LCM = 21: 56/21 + 6/21

= **62/21 = 2 20/21**

1 (k) **Add using Brahmagupta's method: 3/4 + 1/3 + 1/5**

✔ ANSWER

LCM = 60: 45/60 + 20/60 + 12/60

= **77/60 = 1 17/60**

1 (l) **Add using Brahmagupta's method: 2/3 + 4/5 + 3/7**

✔ ANSWER

LCM = 105: 70/105 + 84/105 + 45/105

= **199/105 = 1 94/105**

1 (m) **Add using Brahmagupta's method: 9/2 + 5/4 + 7/6**

✔ ANSWER

LCM = 12: 54/12 + 15/12 + 14/12

= **83/12 = 6 11/12**

2 **Rahim mixes 2/3 litres of yellow paint with 3/4 litres of blue paint to make green paint. What is the volume of green paint he has made?**

✔ ANSWER

2/3 + 3/4

LCM = 12: 8/12 + 9/12 = 17/12

17/12 litres = 1 5/12 litres

3 **Geeta bought 2/5 metre of lace and Shamim bought 3/4 metre of the same lace to put a complete border on a table cloth whose perimeter is 1 metre long. Find the total length of the lace they both have bought. Will the lace be sufficient to cover the whole border?**

✔ ANSWER

2/5 + 3/4

LCM = 20: 8/20 + 15/20 = 23/20

Total = **23/20 m = 1 3/20 m**

23/20 > 20/20 = 1 m, so **yes, the lace is sufficient** (3/20 m extra).

Figure it Out · 7.8 Subtraction with the same fractional unit

1 **5/8 − 3/8**

✔ ANSWER

(5 − 3)/8 = 2/8

= **2/8 = 1/4**

2 **7/9 − 5/9**

✔ ANSWER

(7 − 5)/9

= **2/9**

3 **10/27 − 1/27**

✔ ANSWER

(10 − 1)/27 = 9/27

= **9/27 = 1/3**

Figure it Out · 7.8 Subtraction with different fractional units

1 (a) **8/15 − 3/15**

✔ ANSWER

Same denominator: 5/15

= **1/3**

1 (b) **2/5 − 4/15**

✔ ANSWER

LCM = 15: 6/15 − 4/15

= **2/15**

1 (c) **5/6 − 4/9**

✔ ANSWER

LCM = 18: 15/18 − 8/18

= **7/18**

1 (d) **2/3 − 1/2**

✔ ANSWER

LCM = 6: 4/6 − 3/6

= **1/6**

2 (a) **Subtract 13/4 from 10/3.**

✔ ANSWER

10/3 − 13/4

LCM = 12: 40/12 − 39/12

= **1/12**

2 (b) **Subtract 18/5 from 23/3.**

✔ ANSWER

23/3 − 18/5

LCM = 15: 115/15 − 54/15

= **61/15 = 4 1/15**

2 (c) **Subtract 29/7 from 45/7.**

✔ ANSWER

45/7 − 29/7 = 16/7

= **16/7 = 2 2/7**

3 (a) **Jaya's school is 7/10 km from her home. She takes an auto for 1/2 km and walks the**

remaining distance. How much does she walk daily to reach the school?

✔ ANSWER

Walk = 7/10 − 1/2

= 7/10 − 5/10 = 2/10

= **1/5 km**

3 (b) **Jeevika takes 10/3 minutes to take a complete round of the park and her friend Namit takes 13/4 minutes. Who takes less time and by how much?**

✔ ANSWER

LCM = 12: 10/3 = 40/12, 13/4 = 39/12

39/12 < 40/12, so Namit takes less time.

Difference = 40/12 − 39/12 = 1/12

Namit takes less time, by 1/12 minute.

⭐ 7.9 A Pinch of History · Puzzle!

1 **Can you find three different fractional units that add up to 1?**

✔ ANSWER

Start with 1/3 + 1/3 + 1/3 = 1. One unit must become bigger, so it must be 1/2.

Then 1/2 + 1/4 + 1/4 = 1. One 1/4 must become bigger (not 1/2), so it must be 1/3.

Third fraction = 1 − 1/2 − 1/3 = 6/6 − 3/6 − 2/6 = 1/6

1/2 + 1/3 + 1/6 = 1 (the only solution)

2 **Can you find four different fractional units that add up to 1? (There are six solutions.)**

✔ ANSWER

All six solutions:

- 1/2 + 1/3 + 1/7 + 1/42 = 1
- 1/2 + 1/3 + 1/8 + 1/24 = 1
- 1/2 + 1/3 + 1/9 + 1/18 = 1
- 1/2 + 1/3 + 1/10 + 1/15 = 1
- 1/2 + 1/4 + 1/5 + 1/20 = 1
- 1/2 + 1/4 + 1/6 + 1/12 = 1

Check one: 1/2 + 1/4 + 1/6 + 1/12 = 6/12 + 3/12 + 2/12 + 1/12 = 12/12 = **1**

🎉 **Chapter 7 done! Revise the answers once more tomorrow.**